

WORKED EXAMPLE

One batch means one common starting state

Continue the same 2-to-2-to-2-to-1 sigmoid network from Lessons16-19.

$$A: x = (1, 2), t = 1; \quad B: x = (-1, 1), t = 0$$

$$J = \frac{\ell_A + \ell_B}{2}, \quad \eta = 0.1$$

layer	weight rows (source to destination)	bias vector
1	(0.1, -0.2); (0.3, 0.2)	(0, 0.1)
2	(0.4, -0.3); (-0.2, 0.2)	(0.1, -0.1)
3	(0.3); (-0.4)	(0.2)

These are the original study parameters. Calculate both examples here, average their gradients, and only then change any parameter. All fifteen parameters, including biases, will be updated.

Writing the updates one at a time is fine: simultaneous means every gradient came from this same original state.

WORKED EXAMPLE

Per-example loss first; mean loss second

A subscript A or B identifies an example, not a neuron.

$$\ell_n = -[t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

$$\delta_n^{(3)} = \frac{\partial \ell_n}{\partial z_n^{(3)}} = y_n - t_n$$

$$\frac{\partial J}{\partial \theta} = \frac{1}{2} \left(\frac{\partial \ell_A}{\partial \theta} + \frac{\partial \ell_B}{\partial \theta} \right)$$

Theta here means any one weight or bias. The delta is a derivative of one example loss, so it does not contain the factor 1/2. Average each parameter gradient exactly once.

The assignment writes a mean objective and the y minus target shortcut. We make the per-example meaning explicit before averaging. If you instead define a delta from J, its factor 1/2 is already included.

WORKED EXAMPLE

Original forward values and starting cost

Lesson17 shows all five neuron calculations for both inputs.

quantity	A	B
hidden1 y	0.6681877722, 0.5744425168	0.5498339973, 0.6224593312
hidden2 y	0.5627638384, 0.4537407133	0.5487054962, 0.4649430331
output y	0.5466967333	0.5445402310
BCE loss	0.6038610484	0.7864478889

$$J_{old} = \frac{0.6038610484 + 0.7864478889}{2} = 0.6951544686$$

A gradients were derived in Lessons18-19. Now derive B at the same state. Displayed decimals are approximate; retain full precision in your calculator.

WORKED EXAMPLE

B: start backward at the output

The target is zero, so the output delta is positive.

$$\delta_B^{(3)} = 0.5445402310 - 0 = 0.5445402310$$

For each hidden neuron, sum every outgoing weight times its destination delta. Then multiply by its local sigmoid slope $y(1-y)$.

$$\delta_j^{(l)} = \left(\sum_k w_{j,k}^{(l+1)} \delta_k^{(l+1)} \right) y_j^{(l)} (1 - y_j^{(l)})$$

Use B activations with B deltas. An A activation paired with a B delta would not be the gradient of either example.

WORKED EXAMPLE

B: hidden layer 2, neuron 1 delta

All quantities on this page belong to example B.

$$u = (0.3)(0.5445402310)$$

$$= (0.1633620693) = 0.1633620693$$

$$y(1 - y) = 0.5487054962(1 - 0.5487054962) = 0.2476277746$$

$$\delta_{B,1}^{(2)} = (0.1633620693)(0.2476277746) = 0.0404529857$$

The symbol u is the incoming backward slope from downstream neurons. Add the paths before applying this neuron local slope.

WORKED EXAMPLE

B: hidden layer 2, neuron 2 delta

All quantities on this page belong to example B.

$$u = (-0.4)(0.5445402310)$$

$$= (-0.2178160924) = -0.2178160924$$

$$y(1 - y) = 0.4649430331(1 - 0.4649430331) = 0.2487710091$$

$$\delta_{B,2}^{(2)} = (-0.2178160924)(0.2487710091) = -0.0541863291$$

The symbol u is the incoming backward slope from downstream neurons. Add the paths before applying this neuron local slope.

WORKED EXAMPLE

B: hidden layer 1, neuron 1 delta

All quantities on this page belong to example B.

$$\begin{aligned}u &= (0.4)(0.0404529857) + (-0.3)(-0.0541863291) \\ &= (0.0161811943) + (0.0162558987) = 0.0324370930\end{aligned}$$

$$y(1 - y) = 0.5498339973(1 - 0.5498339973) = 0.2475165727$$

$$\delta_{B,1}^{(1)} = (0.0324370930)(0.2475165727) = 0.0080287181$$

The symbol u is the incoming backward slope from downstream neurons. Add the paths before applying this neuron local slope.

WORKED EXAMPLE

B: hidden layer 1, neuron 2 delta

All quantities on this page belong to example B.

$$\begin{aligned}u &= (-0.2)(0.0404529857) + (0.2)(-0.0541863291) \\&= (-0.0080905971) + (-0.0108372658) = -0.0189278630\end{aligned}$$

$$y(1 - y) = 0.6224593312(1 - 0.6224593312) = 0.2350037122$$

$$\delta_{B,2}^{(1)} = (-0.0189278630)(0.2350037122) = -0.0044481181$$

The symbol u is the incoming backward slope from downstream neurons. Add the paths before applying this neuron local slope.

WORKED EXAMPLE

B: layer 1, neuron 1 gradients

Two weights and one bias: every incoming parameter is included.

weight gradient = source activation × destination delta

$$\frac{\partial \ell_B}{\partial w_{1,1}^{(1)}} = (-1.00000000000)(0.0080287181) = -0.0080287181$$

$$\frac{\partial \ell_B}{\partial w_{2,1}^{(1)}} = (1.00000000000)(0.0080287181) = 0.0080287181$$

$$\frac{\partial \ell_B}{\partial b_1^{(1)}} = 1(0.0080287181) = 0.0080287181$$

The bias has constant source 1. Here the input -1 reverses the first weight-gradient sign.

WORKED EXAMPLE

B: layer 1, neuron 2 gradients

Two weights and one bias: every incoming parameter is included.

weight gradient = source activation × destination delta

$$\frac{\partial \ell_B}{\partial w_{1,2}^{(1)}} = (-1.0000000000)(-0.0044481181) = 0.0044481181$$

$$\frac{\partial \ell_B}{\partial w_{2,2}^{(1)}} = (1.0000000000)(-0.0044481181) = -0.0044481181$$

$$\frac{\partial \ell_B}{\partial b_2^{(1)}} = 1(-0.0044481181) = -0.0044481181$$

The bias has constant source 1. Here the input -1 reverses the first weight-gradient sign.

WORKED EXAMPLE

B: layer 2, neuron 1 gradients

Two weights and one bias: every incoming parameter is included.

weight gradient = source activation × destination delta

$$\frac{\partial \ell_B}{\partial w_{1,1}^{(2)}} = (0.5498339973)(0.0404529857) = 0.0222424268$$

$$\frac{\partial \ell_B}{\partial w_{2,1}^{(2)}} = (0.6224593312)(0.0404529857) = 0.0251803384$$

$$\frac{\partial \ell_B}{\partial b_1^{(2)}} = 1(0.0404529857) = 0.0404529857$$

The bias has constant source 1. Ordinary weight gradients use the previous hidden layer activations shown in the products.

WORKED EXAMPLE

B: layer 2, neuron 2 gradients

Two weights and one bias: every incoming parameter is included.

weight gradient = source activation × destination delta

$$\frac{\partial \ell_B}{\partial w_{1,2}^{(2)}} = (0.5498339973)(-0.0541863291) = -0.0297934859$$

$$\frac{\partial \ell_B}{\partial w_{2,2}^{(2)}} = (0.6224593312)(-0.0541863291) = -0.0337287862$$

$$\frac{\partial \ell_B}{\partial b_2} = 1(-0.0541863291) = -0.0541863291$$

The bias has constant source 1. Ordinary weight gradients use the previous hidden layer activations shown in the products.

WORKED EXAMPLE

B: layer 3, neuron 1 gradients

Two weights and one bias: every incoming parameter is included.

weight gradient = source activation × destination delta

$$\frac{\partial \ell_B}{\partial w_{1,1}^{(3)}} = (0.5487054962)(0.5445402310) = 0.2987922177$$

$$\frac{\partial \ell_B}{\partial w_{2,1}^{(3)}} = (0.4649430331)(0.5445402310) = 0.2531801866$$

$$\frac{\partial \ell_B}{\partial b_1^{(3)}} = 1(0.5445402310) = 0.5445402310$$

The bias has constant source 1. Ordinary weight gradients use the previous hidden layer activations shown in the products.

WORKED EXAMPLE

update layer 1, neuron 1

Learning rate 0.1; the three incoming parameters are updated separately.

$$g_{w_{1,1}^{(1)}} = \frac{(-0.00595687) + (-0.00802872)}{2} = -0.00699280$$

$$(w_{1,1}^{(1)})_{new} = 0.1 - 0.1(-0.00699280) = 0.10069928$$

$$g_{w_{2,1}^{(1)}} = \frac{(-0.01191374) + (0.00802872)}{2} = -0.00194251$$

$$(w_{2,1}^{(1)})_{new} = 0.3 - 0.1(-0.00194251) = 0.30019425$$

$$g_{b_1^{(1)}} = \frac{(-0.00595687) + (0.00802872)}{2} = 0.00103592$$

$$(b_1^{(1)})_{new} = 0 - 0.1(0.00103592) = -0.00010359$$

A negative gradient increases its parameter: subtracting a negative adds.

WORKED EXAMPLE

update layer 1, neuron 2

Learning rate 0.1; the three incoming parameters are updated separately.

$$g_{w_{1,2}}^{(1)} = \frac{(0.00383332) + (0.00444812)}{2} = 0.00414072$$

$$(w_{1,2}^{(1)})_{new} = -0.2 - 0.1(0.00414072) = -0.20041407$$

$$g_{w_{2,2}}^{(1)} = \frac{(0.00766664) + (-0.00444812)}{2} = 0.00160926$$

$$(w_{2,2}^{(1)})_{new} = 0.2 - 0.1(0.00160926) = 0.19983907$$

$$g_{b_2}^{(1)} = \frac{(0.00383332) + (-0.00444812)}{2} = -0.00030740$$

$$(b_2^{(1)})_{new} = 0.1 - 0.1(-0.00030740) = 0.10003074$$

A negative gradient increases its parameter: subtracting a negative adds.

WORKED EXAMPLE

update layer 2, neuron 1

Learning rate 0.1; the three incoming parameters are updated separately.

$$g_{w_{1,1}^{(2)}} = \frac{(-0.02235892) + (0.02224243)}{2} = -0.00005825$$

$$(w_{1,1}^{(2)})_{new} = 0.4 - 0.1(-0.00005825) = 0.40000582$$

$$g_{w_{2,1}^{(2)}} = \frac{(-0.01922202) + (0.02518034)}{2} = 0.00297916$$

$$(w_{2,1}^{(2)})_{new} = -0.2 - 0.1(0.00297916) = -0.20029792$$

$$g_{b_1^{(2)}} = \frac{(-0.03346204) + (0.04045299)}{2} = 0.00349547$$

$$(b_1^{(2)})_{new} = 0.1 - 0.1(0.00349547) = 0.09965045$$

A negative gradient increases its parameter: subtracting a negative adds.

WORKED EXAMPLE

update layer 2, neuron 2

Learning rate 0.1; the three incoming parameters are updated separately.

$$g_{w_{1,2}^{(2)}} = \frac{(0.03002990) + (-0.02979349)}{2} = 0.00011821$$

$$(w_{1,2}^{(2)})_{new} = -0.3 - 0.1(0.00011821) = -0.30001182$$

$$g_{w_{2,2}^{(2)}} = \frac{(0.02581678) + (-0.03372879)}{2} = -0.00395601$$

$$(w_{2,2}^{(2)})_{new} = 0.2 - 0.1(-0.00395601) = 0.20039560$$

$$g_{b_2^{(2)}} = \frac{(0.04494231) + (-0.05418633)}{2} = -0.00462201$$

$$(b_2^{(2)})_{new} = -0.1 - 0.1(-0.00462201) = -0.09953780$$

A negative gradient increases its parameter: subtracting a negative adds.

WORKED EXAMPLE

update layer 3, neuron 1

Learning rate 0.1; the three incoming parameters are updated separately.

$$g_{w_{1,1}^{(3)}} = \frac{(-0.25510269) + (0.29879222)}{2} = 0.02184477$$

$$(w_{1,1}^{(3)})_{new} = 0.3 - 0.1(0.02184477) = 0.29781552$$

$$g_{w_{2,1}^{(3)}} = \frac{(-0.20568215) + (0.25318019)}{2} = 0.02374902$$

$$(w_{2,1}^{(3)})_{new} = -0.4 - 0.1(0.02374902) = -0.40237490$$

$$g_{b_1^{(3)}} = \frac{(-0.45330327) + (0.54454023)}{2} = 0.04561848$$

$$(b_1^{(3)})_{new} = 0.2 - 0.1(0.04561848) = 0.19543815$$

A negative gradient increases its parameter: subtracting a negative adds.

WORKED EXAMPLE

fresh A, layer 1

All fifteen parameters have now changed using learning rate 0.1.

$$z_1^{(1)} = (0.100699)(1.000000) + (0.300194)(2.000000) + (-0.000104)$$

$$= 0.7009841899, \quad y_1^{(1)} = \frac{1}{1 + e^{-(0.7009841899)}} = 0.6684059436$$

$$z_2^{(1)} = (-0.200414)(1.000000) + (0.199839)(2.000000) + (0.100031)$$

$$= 0.2992948161, \quad y_2^{(1)} = \frac{1}{1 + e^{-(0.2992948161)}} = 0.5742701197$$

Recompute from the new parameters and the new previous-layer activations. Reusing an old hidden activation would mix two network states.

WORKED EXAMPLE

fresh A, layer 2

All fifteen parameters have now changed using learning rate 0.1.

$$z_1^{(2)} = (0.400006)(0.668406) + (-0.200298)(0.574270) + (0.099650)$$

$$= 0.2519916150, \quad y_1^{(2)} = \frac{1}{1 + e^{-(0.2519916150)}} = 0.5626666444$$

$$z_2^{(2)} = (-0.300012)(0.668406) + (0.200396)(0.574270) + (-0.099538)$$

$$= -0.1849862780, \quad y_2^{(2)} = \frac{1}{1 + e^{-(-0.1849862780)}} = 0.4538848603$$

Recompute from the new parameters and the new previous-layer activations. Reusing an old hidden activation would mix two network states.

WORKED EXAMPLE

fresh A, layer 3

All fifteen parameters have now changed using learning rate 0.1.

$$z_1^{(3)} = (0.297816)(0.562667) + (-0.402375)(0.453885) + (0.195438)$$

$$= 0.1803771368, \quad y_1^{(3)} = \frac{1}{1 + e^{-(0.1803771368)}} = 0.5449724154$$

$$\ell_{A, new} = -\ln(0.5449724154) = 0.6070200995$$

Recompute from the new parameters and the new previous-layer activations. Reusing an old hidden activation would mix two network states.

WORKED EXAMPLE

fresh B, layer 1

All fifteen parameters have now changed using learning rate 0.1.

$$z_1^{(1)} = (0.100699)(-1.000000) + (0.300194)(1.000000) + (-0.000104)$$

$$= 0.1993913795, \quad y_1^{(1)} = \frac{1}{1 + e^{-(0.1993913795)}} = 0.5496833491$$

$$z_2^{(1)} = (-0.200414)(-1.000000) + (0.199839)(1.000000) + (0.100031)$$

$$= 0.5002838858, \quad y_2^{(1)} = \frac{1}{1 + e^{-(0.5002838858)}} = 0.6225260431$$

Recompute from the new parameters and the new previous-layer activations. Reusing an old hidden activation would mix two network states.

WORKED EXAMPLE

fresh B, layer 2

All fifteen parameters have now changed using learning rate 0.1.

$$z_1^{(2)} = (0.400006)(0.549683) + (-0.200298)(0.622526) + (0.099650)$$

$$= 0.1948363248, \quad y_1^{(2)} = \frac{1}{1 + e^{-(0.1948363248)}} = 0.5485555760$$

$$z_2^{(2)} = (-0.300012)(0.549683) + (0.200396)(0.622526) + (-0.099538)$$

$$= -0.1396978214, \quad y_2^{(2)} = \frac{1}{1 + e^{-(-0.1396978214)}} = 0.4651322313$$

Recompute from the new parameters and the new previous-layer activations. Reusing an old hidden activation would mix two network states.

WORKED EXAMPLE

fresh B, layer 3

All fifteen parameters have now changed using learning rate 0.1.

$$z_1^{(3)} = (0.297816)(0.548556) + (-0.402375)(0.465132) + (0.195438)$$

$$= 0.1716489818, \quad y_1^{(3)} = \frac{1}{1 + e^{-(0.1716489818)}} = 0.5428071933$$

$$\ell_{B, new} = -\ln(1 - 0.5428071933) = 0.7826500806$$

Recompute from the new parameters and the new previous-layer activations. Reusing an old hidden activation would mix two network states.

WORKED EXAMPLE

compare the new batch mean

The training objective is the average over both examples.

$$J_{new} = \frac{0.6070200995 + 0.7826500806}{2} = 0.6948350901$$

$$J_{new} - J_{old} = 0.6948350901 - 0.6951544686 = -0.0003193785$$

The mean decreases for this chosen step. An individual example loss can increase: gradient descent follows the mean gradient, not a separate best direction for every example.

A decrease here is a numerical check, not a guarantee for every possible learning rate. The calculations retain full precision internally; displayed decimals are rounded.

INDEPENDENT PRACTICE

Your turn: a fresh complete training step

Reset the original fifteen parameters; these are new input examples.

$$C: x = (0, 1), t = 1; \quad D: x = (1, -1), t = 0$$

$$J = (\ell_C + \ell_D)/2, \quad \eta = 0.1$$

Calculate both complete forward passes, losses, all five deltas per example, and all fifteen parameter gradients per example. Average each gradient once, update every parameter, then recompute both complete forwards and the new mean loss.

Explain which C gradients must be zero and why. Explain how D negative second input affects its first-layer weight gradients. Do not reuse A/B gradients: the inputs have changed.

Stop here and attempt the problem before reading the following full worked answers.

WORKED ANSWER

C: original forward, layer 1

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$z_1^{(1)} = (0.100000)(0.000000) + (0.300000)(1.000000) + (0.000000)$$

$$= 0.3000000000, \quad y_1^{(1)} = \frac{1}{1 + e^{-(0.3000000000)}} = 0.5744425168$$

$$z_2^{(1)} = (-0.200000)(0.000000) + (0.200000)(1.000000) + (0.100000)$$

$$= 0.3000000000, \quad y_2^{(1)} = \frac{1}{1 + e^{-(0.3000000000)}} = 0.5744425168$$

Use this newly computed activation in the next layer; keep all digits in a calculator until the final answer.

WORKED ANSWER

C: original forward, layer 2

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$z_1^{(2)} = (0.400000)(0.574443) + (-0.200000)(0.574443) + (0.100000)$$

$$= 0.2148885034, \quad y_1^{(2)} = \frac{1}{1 + e^{-(0.2148885034)}} = 0.5535163485$$

$$z_2^{(2)} = (-0.300000)(0.574443) + (0.200000)(0.574443) + (-0.100000)$$

$$= -0.1574442517, \quad y_2^{(2)} = \frac{1}{1 + e^{-(-0.1574442517)}} = 0.4607200451$$

Use this newly computed activation in the next layer; keep all digits in a calculator until the final answer.

WORKED ANSWER

C: original forward, layer 3

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$z_1^{(3)} = (0.300000)(0.553516) + (-0.400000)(0.460720) + (0.200000)$$

$$= 0.1817668865, \quad y_1^{(3)} = \frac{1}{1 + e^{-(0.1817668865)}} = 0.5453170204$$

$$\ell_C = -\ln(0.5453170204) = 0.6063879646$$

The target enters the loss here. Keep the probability for BCE; do not replace it with a thresholded class.

WORKED ANSWER

C: output delta

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$\delta_1^{(3)} = y - t = 0.5453170204 - 1 = -0.4546829796$$

This derivative is for one example loss. The mean factor 1/2 enters later, after each parameter gradient has been calculated.

WORKED ANSWER

C: hidden layer 2, neuron 1 delta

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$u = (0.3)(-0.4546829796)$$

$$= (-0.1364048939) = -0.1364048939$$

$$y(1 - y) = 0.5535163485(1 - 0.5535163485) = 0.2471360004$$

$$\delta_1^{(2)} = (-0.1364048939)(0.2471360004) = -0.0337105599$$

Here u is the incoming backward derivative: add all downstream weighted deltas, then multiply by this neuron local sigmoid slope.

WORKED ANSWER

C: hidden layer 2, neuron 2 delta

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$u = (-0.4)(-0.4546829796)$$

$$= (0.1818731918) = 0.1818731918$$

$$y(1 - y) = 0.4607200451(1 - 0.4607200451) = 0.2484570851$$

$$\delta_2^{(2)} = (0.1818731918)(0.2484570851) = 0.0451876831$$

Here u is the incoming backward derivative: add all downstream weighted deltas, then multiply by this neuron local sigmoid slope.

WORKED ANSWER

C: hidden layer 1, neuron 1 delta

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$u = (0.4)(-0.0337105599) + (-0.3)(0.0451876831)$$

$$= (-0.0134842240) + (-0.0135563049) = -0.0270405289$$

$$y(1 - y) = 0.5744425168(1 - 0.5744425168) = 0.2444583117$$

$$\delta_1^{(1)} = (-0.0270405289)(0.2444583117) = -0.0066102820$$

Here u is the incoming backward derivative: add all downstream weighted deltas, then multiply by this neuron local sigmoid slope.

WORKED ANSWER

C: hidden layer 1, neuron 2 delta

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$u = (-0.2)(-0.0337105599) + (0.2)(0.0451876831)$$

$$= (0.0067421120) + (0.0090375366) = 0.0157796486$$

$$y(1 - y) = 0.5744425168(1 - 0.5744425168) = 0.2444583117$$

$$\delta_2^{(1)} = (0.0157796486)(0.2444583117) = 0.0038574663$$

Here u is the incoming backward derivative: add all downstream weighted deltas, then multiply by this neuron local sigmoid slope.

WORKED ANSWER

C: layer 1, neuron 1 gradients

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{i,j}^{(l)}} = y_i^{(l-1)} \delta_j^{(l)}$$

$$g_{w_{1,1}^{(1)}} = (0.0000000000)(-0.0066102820) = 0.0000000000$$

$$g_{w_{2,1}^{(1)}} = (1.0000000000)(-0.0066102820) = -0.0066102820$$

$$g_{b_1^{(1)}} = 1(-0.0066102820) = -0.0066102820$$

C first input is zero, so both first-input weight gradients are zero, even though the destination deltas are nonzero.

WORKED ANSWER

C: layer 1, neuron 2 gradients

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{i,j}^{(l)}} = y_i^{(l-1)} \delta_j^{(l)}$$

$$g_{w_{1,2}^{(1)}} = (0.0000000000)(0.0038574663) = 0.0000000000$$

$$g_{w_{2,2}^{(1)}} = (1.0000000000)(0.0038574663) = 0.0038574663$$

$$g_{b_2^{(1)}} = 1(0.0038574663) = 0.0038574663$$

C first input is zero, so both first-input weight gradients are zero, even though the destination deltas are nonzero.

WORKED ANSWER

C: layer 2, neuron 1 gradients

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{i,j}^{(l)}} = y_i^{(l-1)} \delta_j^{(l)}$$

$$g_{w_{1,1}^{(2)}} = (0.5744425168)(-0.0337105599) = -0.0193647789$$

$$g_{w_{2,1}^{(2)}} = (0.5744425168)(-0.0337105599) = -0.0193647789$$

$$g_{b_1^{(2)}} = 1(-0.0337105599) = -0.0337105599$$

Each weight uses its own source activation; the bias uses the constant source 1.

WORKED ANSWER

C: layer 2, neuron 2 gradients

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{i,j}^{(l)}} = y_i^{(l-1)} \delta_j^{(l)}$$

$$g_{w_{1,2}^{(2)}} = (0.5744425168)(0.0451876831) = 0.0259577264$$

$$g_{w_{2,2}^{(2)}} = (0.5744425168)(0.0451876831) = 0.0259577264$$

$$g_{b_2^{(2)}} = 1(0.0451876831) = 0.0451876831$$

Each weight uses its own source activation; the bias uses the constant source 1.

WORKED ANSWER

C: layer 3, neuron 1 gradients

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{i,j}^{(l)}} = y_i^{(l-1)} \delta_j^{(l)}$$

$$g_{w_{1,1}^{(3)}} = (0.5535163485)(-0.4546829796) = -0.2516744626$$

$$g_{w_{2,1}^{(3)}} = (0.4607200451)(-0.4546829796) = -0.2094815629$$

$$g_{b_1^{(3)}} = 1(-0.4546829796) = -0.4546829796$$

Each weight uses its own source activation; the bias uses the constant source 1.

WORKED ANSWER

D: original forward, layer 1

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$z_1^{(1)} = (0.100000)(1.000000) + (0.300000)(-1.000000) + (0.000000)$$

$$= -0.2000000000, \quad y_1^{(1)} = \frac{1}{1 + e^{-(-0.2000000000)}} = 0.4501660027$$

$$z_2^{(1)} = (-0.200000)(1.000000) + (0.200000)(-1.000000) + (0.100000)$$

$$= -0.3000000000, \quad y_2^{(1)} = \frac{1}{1 + e^{-(-0.3000000000)}} = 0.4255574832$$

Use this newly computed activation in the next layer; keep all digits in a calculator until the final answer.

WORKED ANSWER

D: original forward, layer 2

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$z_1^{(2)} = (0.400000)(0.450166) + (-0.200000)(0.425557) + (0.100000)$$

$$= 0.1949549044, \quad y_1^{(2)} = \frac{1}{1 + e^{-(0.1949549044)}} = 0.5485849412$$

$$z_2^{(2)} = (-0.300000)(0.450166) + (0.200000)(0.425557) + (-0.100000)$$

$$= -0.1499383042, \quad y_2^{(2)} = \frac{1}{1 + e^{-(-0.1499383042)}} = 0.4625854922$$

Use this newly computed activation in the next layer; keep all digits in a calculator until the final answer.

WORKED ANSWER

D: original forward, layer 3

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$z_1^{(3)} = (0.300000)(0.548585) + (-0.400000)(0.462585) + (0.200000)$$

$$= 0.1795412855, \quad y_1^{(3)} = \frac{1}{1 + e^{-(0.1795412855)}} = 0.5447651353$$

$$\ell_D = -\ln(1 - 0.5447651353) = 0.7869418071$$

The target enters the loss here. Keep the probability for BCE; do not replace it with a thresholded class.

WORKED ANSWER

D: output delta

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$\delta_1^{(3)} = y - t = 0.5447651353 - 0 = 0.5447651353$$

This derivative is for one example loss. The mean factor 1/2 enters later, after each parameter gradient has been calculated.

WORKED ANSWER

D: hidden layer 2, neuron 1 delta

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$u = (0.3)(0.5447651353)$$

$$= (0.1634295406) = 0.1634295406$$

$$y(1 - y) = 0.5485849412(1 - 0.5485849412) = 0.2476395035$$

$$\delta_1^{(2)} = (0.1634295406)(0.2476395035) = 0.0404716103$$

Here u is the incoming backward derivative: add all downstream weighted deltas, then multiply by this neuron local sigmoid slope.

WORKED ANSWER

D: hidden layer 2, neuron 2 delta

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$u = (-0.4)(0.5447651353)$$

$$= (-0.2179060541) = -0.2179060541$$

$$y(1 - y) = 0.4625854922(1 - 0.4625854922) = 0.2486001546$$

$$\delta_2^{(2)} = (-0.2179060541)(0.2486001546) = -0.0541714787$$

Here u is the incoming backward derivative: add all downstream weighted deltas, then multiply by this neuron local sigmoid slope.

WORKED ANSWER

D: hidden layer 1, neuron 1 delta

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$u = (0.4)(0.0404716103) + (-0.3)(-0.0541714787)$$

$$= (0.0161886441) + (0.0162514436) = 0.0324400877$$

$$y(1 - y) = 0.4501660027(1 - 0.4501660027) = 0.2475165727$$

$$\delta_1^{(1)} = (0.0324400877)(0.2475165727) = 0.0080294593$$

Here u is the incoming backward derivative: add all downstream weighted deltas, then multiply by this neuron local sigmoid slope.

WORKED ANSWER

D: hidden layer 1, neuron 2 delta

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$\begin{aligned}u &= (-0.2)(0.0404716103) + (0.2)(-0.0541714787) \\&= (-0.0080943221) + (-0.0108342957) = -0.0189286178\end{aligned}$$

$$y(1 - y) = 0.4255574832(1 - 0.4255574832) = 0.2444583117$$

$$\delta_2^{(1)} = (-0.0189286178)(0.2444583117) = -0.0046272580$$

Here u is the incoming backward derivative: add all downstream weighted deltas, then multiply by this neuron local sigmoid slope.

WORKED ANSWER

D: layer 1, neuron 1 gradients

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{i,j}^{(l)}} = y_i^{(l-1)} \delta_j^{(l)}$$

$$g_{w_{1,1}^{(1)}} = (1.0000000000)(0.0080294593) = 0.0080294593$$

$$g_{w_{2,1}^{(1)}} = (-1.0000000000)(0.0080294593) = -0.0080294593$$

$$g_{b_1^{(1)}} = 1(0.0080294593) = 0.0080294593$$

D second input is -1, so its second-input gradients have the opposite sign from the destination delta.

WORKED ANSWER

D: layer 1, neuron 2 gradients

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{i,j}^{(l)}} = y_i^{(l-1)} \delta_j^{(l)}$$

$$g_{w_{1,2}^{(1)}} = (1.0000000000)(-0.0046272580) = -0.0046272580$$

$$g_{w_{2,2}^{(1)}} = (-1.0000000000)(-0.0046272580) = 0.0046272580$$

$$g_{b_2^{(1)}} = 1(-0.0046272580) = -0.0046272580$$

D second input is -1, so its second-input gradients have the opposite sign from the destination delta.

WORKED ANSWER

D: layer 2, neuron 1 gradients

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{i,j}^{(l)}} = y_i^{(l-1)} \delta_j^{(l)}$$

$$g_{w_{1,1}^{(2)}} = (0.4501660027)(0.0404716103) = 0.0182189430$$

$$g_{w_{2,1}^{(2)}} = (0.4255574832)(0.0404716103) = 0.0172229966$$

$$g_{b_1^{(2)}} = 1(0.0404716103) = 0.0404716103$$

Each weight uses its own source activation; the bias uses the constant source 1.

WORKED ANSWER

D: layer 2, neuron 2 gradients

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{i,j}^{(l)}} = y_i^{(l-1)} \delta_j^{(l)}$$

$$g_{w_{1,2}^{(2)}} = (0.4501660027)(-0.0541714787) = -0.0243861580$$

$$g_{w_{2,2}^{(2)}} = (0.4255574832)(-0.0541714787) = -0.0230530782$$

$$g_{b_2^{(2)}} = 1(-0.0541714787) = -0.0541714787$$

Each weight uses its own source activation; the bias uses the constant source 1.

WORKED ANSWER

D: layer 3, neuron 1 gradients

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{i,j}^{(l)}} = y_i^{(l-1)} \delta_j^{(l)}$$

$$g_{w_{1,1}^{(3)}} = (0.5485849412)(0.5447651353) = 0.2988499497$$

$$g_{w_{2,1}^{(3)}} = (0.4625854922)(0.5447651353) = 0.2520004483$$

$$g_{b_1^{(3)}} = 1(0.5447651353) = 0.5447651353$$

Each weight uses its own source activation; the bias uses the constant source 1.

WORKED ANSWER

Practice: initial mean loss

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$J_{old} = \frac{0.6063879646 + 0.7869418071}{2} = 0.6966648858$$

We now have both example gradients at one common parameter state. Average C and D for each corresponding parameter, then subtract learning rate times that mean.

WORKED ANSWER

Practice: update layer 1, neuron 1

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{1,1}^{(1)}} = \frac{(0.00000000) + (0.00802946)}{2} = 0.00401473$$

$$(w_{1,1}^{(1)})_{new} = 0.1 - 0.1(0.00401473) = 0.09959853$$

$$g_{w_{2,1}^{(1)}} = \frac{(-0.00661028) + (-0.00802946)}{2} = -0.00731987$$

$$(w_{2,1}^{(1)})_{new} = 0.3 - 0.1(-0.00731987) = 0.30073199$$

$$g_{b_1^{(1)}} = \frac{(-0.00661028) + (0.00802946)}{2} = 0.00070959$$

$$(b_1^{(1)})_{new} = 0 - 0.1(0.00070959) = -0.00007096$$

Average matching gradients; every update starts from the original state.

WORKED ANSWER

Practice: update layer 1, neuron 2

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{1,2}}^{(1)} = \frac{(0.00000000) + (-0.00462726)}{2} = -0.00231363$$

$$(w_{1,2}^{(1)})_{new} = -0.2 - 0.1(-0.00231363) = -0.19976864$$

$$g_{w_{2,2}}^{(1)} = \frac{(0.00385747) + (0.00462726)}{2} = 0.00424236$$

$$(w_{2,2}^{(1)})_{new} = 0.2 - 0.1(0.00424236) = 0.19957576$$

$$g_{b_2}^{(1)} = \frac{(0.00385747) + (-0.00462726)}{2} = -0.00038490$$

$$(b_2^{(1)})_{new} = 0.1 - 0.1(-0.00038490) = 0.10003849$$

Average matching gradients; every update starts from the original state.

WORKED ANSWER

Practice: update layer 2, neuron 1

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{1,1}^{(2)}} = \frac{(-0.01936478) + (0.01821894)}{2} = -0.00057292$$

$$(w_{1,1}^{(2)})_{new} = 0.4 - 0.1(-0.00057292) = 0.40005729$$

$$g_{w_{2,1}^{(2)}} = \frac{(-0.01936478) + (0.01722300)}{2} = -0.00107089$$

$$(w_{2,1}^{(2)})_{new} = -0.2 - 0.1(-0.00107089) = -0.19989291$$

$$g_{b_1^{(2)}} = \frac{(-0.03371056) + (0.04047161)}{2} = 0.00338053$$

$$(b_1^{(2)})_{new} = 0.1 - 0.1(0.00338053) = 0.09966195$$

Average matching gradients; every update starts from the original state.

WORKED ANSWER

Practice: update layer 2, neuron 2

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{1,2}^{(2)}} = \frac{(0.02595773) + (-0.02438616)}{2} = 0.00078578$$

$$(w_{1,2}^{(2)})_{new} = -0.3 - 0.1(0.00078578) = -0.30007858$$

$$g_{w_{2,2}^{(2)}} = \frac{(0.02595773) + (-0.02305308)}{2} = 0.00145232$$

$$(w_{2,2}^{(2)})_{new} = 0.2 - 0.1(0.00145232) = 0.19985477$$

$$g_{b_2^{(2)}} = \frac{(0.04518768) + (-0.05417148)}{2} = -0.00449190$$

$$(b_2^{(2)})_{new} = -0.1 - 0.1(-0.00449190) = -0.09955081$$

Average matching gradients; every update starts from the original state.

WORKED ANSWER

Practice: update layer 3, neuron 1

Fresh practice answer: original parameters, C and D batch, learning rate 0.1.

$$g_{w_{1,1}^{(3)}} = \frac{(-0.25167446) + (0.29884995)}{2} = 0.02358774$$

$$(w_{1,1}^{(3)})_{new} = 0.3 - 0.1(0.02358774) = 0.29764123$$

$$g_{w_{2,1}^{(3)}} = \frac{(-0.20948156) + (0.25200045)}{2} = 0.02125944$$

$$(w_{2,1}^{(3)})_{new} = -0.4 - 0.1(0.02125944) = -0.40212594$$

$$g_{b_1^{(3)}} = \frac{(-0.45468298) + (0.54476514)}{2} = 0.04504108$$

$$(b_1^{(3)})_{new} = 0.2 - 0.1(0.04504108) = 0.19549589$$

Average matching gradients; every update starts from the original state.

WORKED ANSWER

C: updated forward, layer 1

Fresh practice answer: updated parameters after the C/D batch step.

$$z_1^{(1)} = (0.099599)(0.000000) + (0.300732)(1.000000) + (-0.000071)$$

$$= 0.3006610282, \quad y_1^{(1)} = \frac{1}{1 + e^{-(0.3006610282)}} = 0.5746041027$$

$$z_2^{(1)} = (-0.199769)(0.000000) + (0.199576)(1.000000) + (0.100038)$$

$$= 0.2996142534, \quad y_2^{(1)} = \frac{1}{1 + e^{-(0.2996142534)}} = 0.5743482151$$

Use this newly computed activation in the next layer; keep all digits in a calculator until the final answer.

WORKED ANSWER

C: updated forward, layer 2

Fresh practice answer: updated parameters after the C/D batch step.

$$z_1^{(2)} = (0.400057)(0.574604) + (-0.199893)(0.574348) + (0.099662)$$

$$= 0.2147283721, \quad y_1^{(2)} = \frac{1}{1 + e^{-(0.2147283721)}} = 0.5534767739$$

$$z_2^{(2)} = (-0.300079)(0.574604) + (0.199855)(0.574348) + (-0.099551)$$

$$= -0.1571909635, \quad y_2^{(2)} = \frac{1}{1 + e^{-(-0.1571909635)}} = 0.4607829770$$

Use this newly computed activation in the next layer; keep all digits in a calculator until the final answer.

WORKED ANSWER

C: updated forward, layer 3

Fresh practice answer: updated parameters after the C/D batch step.

$$z_1^{(3)} = (0.297641)(0.553477) + (-0.402126)(0.460783) + (0.195496)$$

$$= 0.1749406078, \quad y_1^{(3)} = \frac{1}{1 + e^{-(0.1749406078)}} = 0.5436239523$$

$$\ell_C = -\ln(0.5436239523) = 0.6094975354$$

The target enters the loss here. Keep the probability for BCE; do not replace it with a thresholded class.

WORKED ANSWER

D: updated forward, layer 1

Fresh practice answer: updated parameters after the C/D batch step.

$$z_1^{(1)} = (0.099599)(1.000000) + (0.300732)(-1.000000) + (-0.000071)$$

$$= -0.2012044189, \quad y_1^{(1)} = \frac{1}{1 + e^{-(-0.2012044189)}} = 0.4498679070$$

$$z_2^{(1)} = (-0.199769)(1.000000) + (0.199576)(-1.000000) + (0.100038)$$

$$= -0.2993059113, \quad y_2^{(1)} = \frac{1}{1 + e^{-(-0.2993059113)}} = 0.4257271677$$

Use this newly computed activation in the next layer; keep all digits in a calculator until the final answer.

WORKED ANSWER

D: updated forward, layer 2

Fresh practice answer: updated parameters after the C/D batch step.

$$z_1^{(2)} = (0.400057)(0.449868) + (-0.199893)(0.425727) + (0.099662)$$

$$= 0.1945350412, \quad y_1^{(2)} = \frac{1}{1 + e^{-(0.1945350412)}} = 0.5484809643$$

$$z_2^{(2)} = (-0.300079)(0.449868) + (0.199855)(0.425727) + (-0.099551)$$

$$= -0.1494629281, \quad y_2^{(2)} = \frac{1}{1 + e^{-(-0.1494629281)}} = 0.4627036729$$

Use this newly computed activation in the next layer; keep all digits in a calculator until the final answer.

WORKED ANSWER

D: updated forward, layer 3

Fresh practice answer: updated parameters after the C/D batch step.

$$z_1^{(3)} = (0.297641)(0.548481) + (-0.402126)(0.462704) + (0.195496)$$

$$= 0.1726812873, \quad y_1^{(3)} = \frac{1}{1 + e^{-(0.1726812873)}} = 0.5430633667$$

$$\ell_D = -\ln(1 - 0.5430633667) = 0.7832105557$$

The target enters the loss here. Keep the probability for BCE; do not replace it with a thresholded class.

WORKED ANSWER

Practice: refreshed mean and readiness check

Fresh practice answer: compare the original and updated C/D batch means.

$$I_{new} = \frac{0.6094975354 + 0.7832105557}{2} = 0.6963540456$$

$$\Delta J = 0.6963540456 - 0.6966648858 = -0.0003108403$$

Both examples were evaluated with all fifteen new parameters. Compare your intermediate values to locate the first mismatch, rather than changing later numbers to match the final mean.

Readiness: explain every source activation, every destination delta, every branch sum, the single averaging factor, and why the second forward pass cannot reuse the old cache.

WORKED ANSWER

Checks that prevent common batch mistakes

All fifteen mean gradients were independently checked by finite differences.

$\text{mean}(a_n \delta_n) \neq \text{mean}(a_n) \text{mean}(\delta_n)$ in general

Multiply the source activation and destination delta for each example first, then average those parameter gradients. Do not average the activations and deltas separately.

Do not divide by two twice, update after A before evaluating B, forget the biases, or reuse the old forward cache after updating.

For the main A/B batch, eta 0.1 gives mean 0.6948350901. Its A loss increases even though the mean decreases. Judge this training step by the stated mean objective.

Finite differences perturb one original parameter at a time, recompute both example losses, average, and compare the numerical slope with the calculated mean gradient.